MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 7C23

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode

Outline: See figure 1
Mounting position: Vertical, anode down
Weight: 5½ pounds nominal

ABSOLUTE RATINGS: (see note 1)

Parameter:	Fl	Ef	eb	Ec	Ib	Pp
Unit:	MHz	Vac	kv	Vdc	Adc	W
Maximum:	50	11.55	15	-1500	1.25	1200
Minimum:	---	10.45	---	---	---	---
TEST CONDITIONS:	---	11.0	3.5	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Conditions	AQL	Insp. Level	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1							
1301	Filament current		.65	II	If	27.5	30.5	Aac
1256	Electrode current (grid)	Ec/Ib = 0.35Adc; t = 180 (See note 3)	.65	II	Ic	---	15	μAdc
1261	Electrode voltage (anode)	Ec = 0; Ib = 0.2Adc	.65	II	Eb	1.6	2.1	kVdc
1261	Electrode voltage (anode)	Ec = -50Vdc; Ib = 0.2Adc	.65	II	Eb	2.6	3.2	kVdc
1316	Amplification factor	Ec/Ib = 0.2Adc	.65	II	Mu	20.0	26.5	---
1261	Electrode voltage (grid)	Ec/Ib = 100μAdc; Eb = 15kVdc	.65	II	Ec	---	-1000	Vdc
1231	Peak emission	ec = eb = 4000v (See note 3)	.65	II	is	25	---	a
---	Insulation	Rp = 300; Ib = 4 to 6mAdc; Eb = 20kVdc (see notes 2 and 3)	.65	II	---	---	---	---
---	Power output	Ib = 0.75Adc; Ic = 0.15Adc; Rg = 2,000; F = 2MHz	.65	II	Po	1500	---	W (useful)
	Quality conformance inspection, part 2							
1331	Direct-Interelectrode capacitance		---	---	Cgp	11.4	15.0	pF
			---	---	Cgi	12.6	16.4	pF
			---	---	Cpi	1.2	2.2	pF
	Quality conformance Inspection, part 3							
---	Life-test provisions	Group D, Power output; t = 500 hours	---	---	---	---	---	---
---	Life-test end point	Peak emission	---	---	is	20	---	a

NOTES:

1. Tube may be used under pulse conditions in which anode voltage may be exceeded for certain conditions.
2. Anode potential is slowly increased to 20kVdc at which potential the time without flashovers shall be less than 300 seconds. A capacity of not less than 0.1 mfd shall be connected in parallel with the anode supply.
3. Test shall be performed at the conclusion of the holding period.

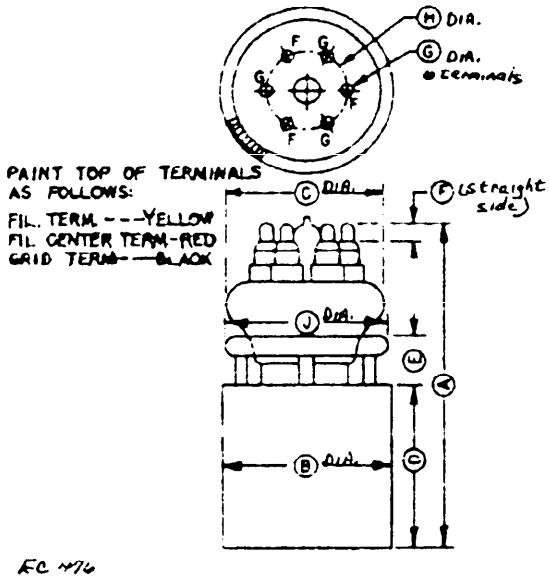
Review activity:
Navy - CG

User activities:
Navy - AS, OS, MC, SH

Preparing activity:
Navy - EC

(Project 5960-N892)

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Ltr	Dimensions			
	Minimum		Maximum	
QC1 part 2				
A	6.50	(165.1)	7.00	(177.8)
B	3.469	(88.11)	3.531	(89.69)
D	3.375	(85.73)	3.625	(92.08)
QC1 part 3 (periodic-check)				
C			3.25	(82.55)
E	.875	(22.23)	1.000	(25.40)
F	.375	(9.53)		
G	.359	(9.12)	.391	(9.93)
H	1.375	(34.93)	1.625	(41.28)
J			3.75	(95.25)

FIGURE 1. Outline drawing of electron tube type 7C23.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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NAVAL ELECTRONIC SYSTEMS COMMAND
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